

SENTINEL: AI-Enhanced Real-Time Traction Control for Autonomous Mobile Robots using μ T-Kernel 3.0

1. Overview and Purpose



In my research on autonomous mobile robots (AMR), I previously developed a DC motor controller using an **Arduino Nano**. However, using Arduino Nano makes the robot less reactive to dynamic environments, often resulting in **skidding**. While I handled this with a PID controller, it wasn't enough to ensure smooth navigation.

To move beyond the limitations of Arduino, I want to replace the low-level controller with **μ T-Kernel 3.0** to maximize the RTOS capabilities. I will optimize sensor handling through multitasking and implement **AI-based Friction Optimization** using **TinyML**. This system will detect skidding from IMU vibrations or current levels before adjusting the DC motors. This project combines a high-level navigation system using **ROS 2 Jazzy**, a deterministic RTOS with **μ T-Kernel 3.0**, and edge AI for traction control.

2. Development/Tools

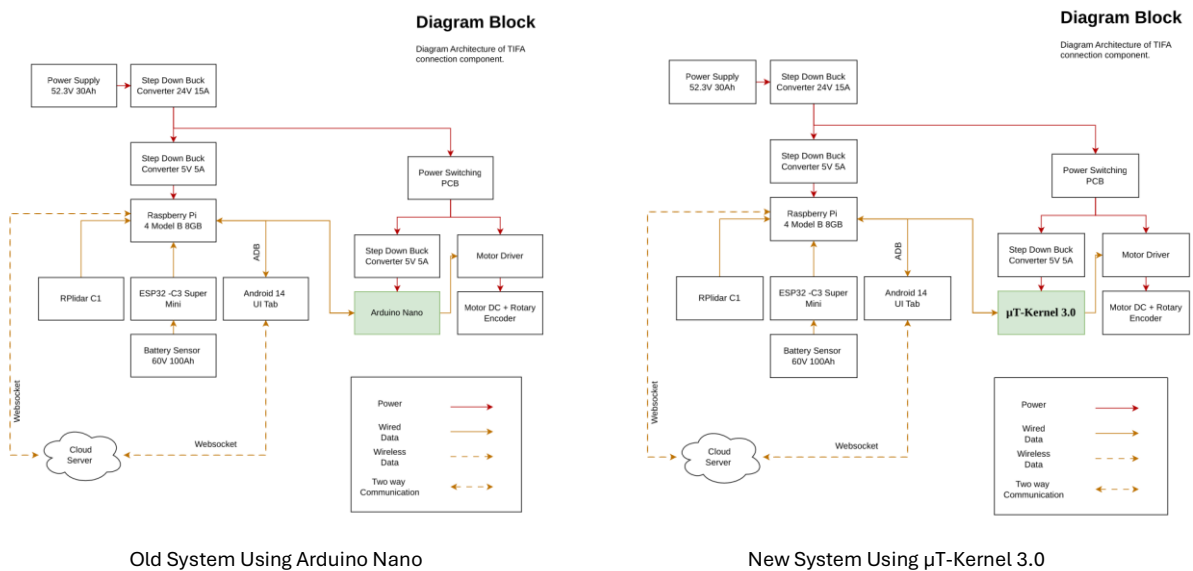
- **OS:** Ubuntu 24.04 LTS (Jazzy/Kilted)
- **Languages:** C / C++ (Low-level), Python (Navigation)
- **IDE:** VS Code
- **Hardware:** Microprocessor Board from TRON committee, Raspberry Pi 5 8GB

3. Roadmap

- **Porting:** Porting μ T-Kernel 3.0 to the board and configuring the multitasking tasks.
- **Low-Level Control:** Implementing sensor reading (Encoder & IMU) and PID logic to send PWM signals to the motor drivers.

- **AI Deployment:** Developing a TinyML model for friction prediction, validating its reliability, and deploying it on the μ T-Kernel board.
- **Integration:** Setting up the Raspberry Pi 5 for SLAM and Navigation, then integrating it with the low-level controller via **UART communication**.

4. Description of Application Program



The application is a **Real-Time Control Hub** that migrates the TIFA robot's "reflex system" from a legacy bare-metal architecture to an industrial RTOS standard.

a. Technical Comparison: Arduino Nano vs. μ T-Kernel 3.0

Based on the provided diagram, the previous system used an **Arduino Nano** which has significant limitations that this program solves:

Feature	Legacy System (Arduino Nano)	Proposed System (μ T-Kernel 3.0)
Architecture	Single-threaded (Sequential Loop)	Preemptive Multitasking (Parallel Tasks)
Task Handling	Motor control can be delayed by sensor reading	Zero-latency priority for PID and Encoder tasks
Reliability	"Frozen" code if one process lags	Independent Task Watchdog for safety
Intelligence	No room for AI/ML processing	Native TinyML Integration for Friction Prediction

b. System Workflow & Data Pipeline

The program manages the components in the diagram with a **Deterministic Task Scheduler**:

- **High-Priority Control Task:** Handles the **Motor Driver** and **Rotary Encoder**. Unlike the Arduino version, the μ T-Kernel ensures the PID loop runs at a constant 1kHz without being interrupted by other processes.
- **AI Friction Optimizer (The "Brain"):** This task analyzes high-frequency vibration data from the **IMU**. When the AI detects a "skidding pattern" (e.g., high RPM but low IMU displacement), it overrides the PWM signal to regain traction.
- **Clean Data Bridge:** The program filters raw sensor noise before transmitting "Clean Odometry" via UART to the **Raspberry Pi 5**. This significantly improves the **SLAM (RPLidar C1)** accuracy compared to the noisy data from the previous Arduino system.

c. Power & Safety Management

The application also monitors the **Battery Sensor** and controls the **Power Switching PCB**. By using the RTOS power-saving features, the board can enter "Low Power Mode" during idle states in the coffee shop, a feature that was not optimized in the previous Arduino-based setup.

5. The Comparison and Uniqueness

In previous developments, TIFA used an Arduino Nano to control the DC motors through a motor driver. This legacy system relied on a traditional, single-task PID method, which couldn't handle advanced AI-based friction optimization. To manage multiple tasks like sensor reading and complex control algorithms, a high-performance board is necessary. Implementing **μ T-Kernel 3.0** allows for a professional RTOS and AI implementation that is much more reliable than the standard Arduino framework.

6. The Passion

As a Master's student in Electrical Engineering at Telkom University, I have the opportunity to further develop **TIFA (Tel-U Interactive Food Assistant)**, a project I started during my Bachelor's degree. TIFA is already serving people at the Telkom University coffee shop. My goal this year is to commercialize TIFA. I lead a team of 10–20 people focusing on electronics, battery systems, and mechanical design. Having led this team for 3 years, I have a big vision: evolving TIFA into a **medical robot assistant**. This RTOS and AI competition is the perfect chance to enhance my product beyond its current limits.